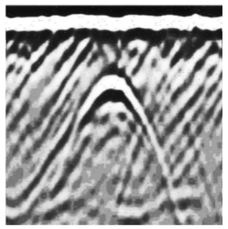
Input
1×640×640



BACKBONE

Block 1
32×640× 640

pool

Block 2
64×320×320

pool

Block 3
128×160×160

pool

Bottleneck
256×80×80

f_3

SUPERVISED ATTENTION BRANCH

Attention head
3×3 conv → BN → ReLU → 1×1 conv

σ

Attention A
1×160×160

AvgPool 2×2
1×80×80

HBAT
params → mask

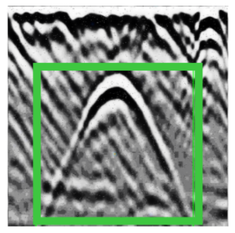
Band mask B
1×160×160

L_{att}
BCE + Dice

Fusion
concat → 1×1 conv
256×80×80

Detection head
obj (1)
size (2)
offset (2)

Prediction



Block = 2 × (3×3 Conv-BN-ReLU);
pool = 2×2 max-pool;
sizes are C×H×W

training only

training and inference